**Date: ……**

**Practical No: 2:** Write a Program to implement Caesar Cipher for basic encryption and decryption.

**A. Objective:**

Develop a Python program to implement the Caesar Cipher for encrypting and decrypting text, showcasing fundamental cryptographic principles.

**B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO5, PO11**

**C. Expected Skills to be developed based on competency:**

The practical is expected to develop the following skills:

- **Algorithmic Implementation**: Develop the ability to implement the Caesar Cipher algorithm effectively, understanding the steps involved in both encryption and decryption.
- **Coding Proficiency in Python**: Enhance Python programming skills by writing code for user input, string manipulation, and modular arithmetic required for the Caesar Cipher.
- **Cryptography Awareness**: Gain an understanding of basic cryptographic concepts, especially in the context of symmetric key encryption, as demonstrated by the Caesar Cipher.
- **Debugging and Troubleshooting**: Cultivate skills in identifying and fixing errors in the code, ensuring the program works correctly for various input scenarios.
- **User Interaction and Input Handling**: Learn to interact with users by handling input for the text to be encrypted or decrypted, providing a practical application of user input processing in programming.

**D. Expected Course Outcomes (Cos): CO2**

**E. Practical Outcome (PRo)**

Write a Program to implement Caesar Cipher for basic encryption and decryption.

**F. Expected Affective domain Outcome (ADos)**

1. Enhanced Confidence in Programming.
2. Developed Problem-Solving Skills.
3. Appreciation for Cryptographic Concepts.
4. Critical Thinking in Security Context

**G. Prerequisite Theory:**

**Symmetric Key Encryption**: Familiarity with the concept of symmetric key cryptography, where the same key is used for both encryption and decryption.

**Basic Python Programming**: Knowledge of basic Python syntax, including variables, data types, input/output, and string manipulation.

**Modular Arithmetic**: Understanding modular arithmetic concepts, specifically modulo operations, which are fundamental to the Caesar Cipher algorithm.

**Cryptography Basics**: Awareness of basic cryptographic terms such as plaintext, ciphertext, encryption, and decryption.

**Caesar Cipher Algorithm**: Knowledge of how the Caesar Cipher works, including the shift operation and its impact on the encryption and decryption processes.

**User Input Handling**: Understanding how to take user input in Python and handle it appropriately for further processing.

**Algorithmic Thinking**: Basic algorithmic thinking skills, including the ability to break down a problem into smaller steps and devise a systematic solution.

**Security Considerations**: An introductory understanding of security concepts, especially the limitations and vulnerabilities of simple encryption methods like the Caesar Cipher.

**Definition and example of Caesar Cipher**

The Caesar Cipher is a simple substitution cipher in cryptography, where each letter in the plaintext is shifted a certain number of positions down or up the alphabet. It is a type of symmetric key algorithm, meaning the same key is used for both encryption and decryption. The cipher is named after Julius Caesar, who is historically known to have used it to encrypt his private correspondence.

Here's a brief explanation of how the Caesar Cipher works with an example:

**Algorithm:**

**Key Generation:**

Choose a shift value, often referred to as the "key." This key determines the amount by which each letter will be shifted.

**Encryption:**

For each letter in the plaintext, shift it by the key value.

Wrap around to the beginning of the alphabet if the shift goes beyond 'Z' (for positive shifts) or to the end if it goes before 'A' (for negative shifts).

Maintain the case of the letters (uppercase or lowercase).

**Decryption:**

Decryption is essentially the reverse process of encryption. For each letter in the ciphertext, shift it back by the key value.

**Example:**

Let's use a Caesar Cipher with a shift of 3:

**Plaintext: "HELLO"**

**Encryption:**

H -> K (shifted 3 positions)

E -> H

L -> O

L -> O

O -> R

**Ciphertext: "KHOOR"**

To decrypt "KHOOR" back to "HELLO," we perform the reverse shift:

**Ciphertext: "KHOOR"**

**Decryption:**

K -> H (shifted back 3 positions)

H -> E

O -> L

O -> L

R -> O

**Plaintext: "HELLO"**

## H. Resources/Equipment Required

| Sr. No. | Instrument/Equipment /Components/Trainer kit | Specification | Quantity |
|---|---|---|---|
| 1 | Hardware: Computer System | Computer (i3-i5 preferable),RAM minimum 2 GB and onwards | As per batch size |
| 2 | Software: Operating System | Windows/ Linux | |
| 3 | Software | **Text Editor**: Notepad (on Windows), TextEdit (on macOS), or gedit (on Linux)<br>**Integrated Development Environment (IDE)**:<br>• PyCharm<br>• Visual Studio Code<br>• IDLE (Python's built-in IDE)<br>• Jupyter Notebook<br>**Python Interpreter** | |

## I. Safety and necessary Precautions followed

- Ensure proper handling of user input to prevent unintended program behavior and encourage students to use small shift values for a clear understanding, avoiding potential errors or misinterpretations.

## J. Procedure to be followed/Source code:

**K. Practical related Quiz.**

1. What type of encryption does the Caesar Cipher represent?

a) Asymmetric Key Encryption
b) Symmetric Key Encryption
c) Public Key Encryption
d) Hybrid Encryption

2. If the shift value for the Caesar Cipher is 3, what will be the encrypted form of the letter 'A'?

a) A
b) B
c) C
d) D

3. In the Caesar Cipher, what happens when a shift goes beyond 'Z' in the alphabet during encryption?

a) It wraps around to 'A'
b) It remains unchanged
c) It is skipped
d) It becomes a space character

4. What is the key consideration when handling user input in the Caesar Cipher program?

a) Ignoring user input
b) Ensuring proper handling to avoid errors
c) Using complex shift values
d) Converting all input to uppercase

5. If the plaintext is "PYTHON" and the shift is 2, what is the corresponding ciphertext?

a) QVJGTQ
b) RWKPVQ
c) RJCUPR
d) RVJGTQ

**M. References / Suggestions ( lab manual designer should give)**

1. https://www.javatpoint.com/caesar-cipher-technique
2. https://www.geeksforgeeks.org/caesar-cipher-in-cryptography/
3. https://www.youtube.com/watch?v=JtbKh_12ctg